Class Test 2

Electricity and Electronics EBN 111

31 May 2021

Test Information			
Maximum marks:	16	Full marks:	15
Duration of test:	35 min	Open/Closed book:	Open
Total number of pages (this page included)		8	

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Important

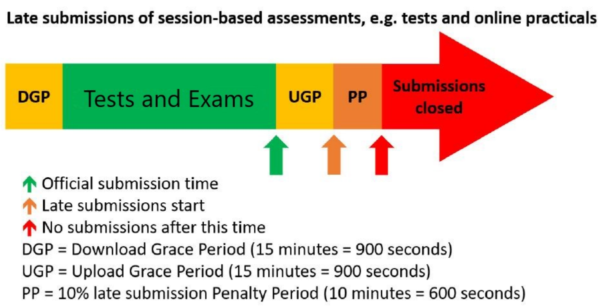
1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write irrational numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By submitting your class test, you declare the following:**

   The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, examinations and/or any other forms of assessment.

Online assessment submission rules and time-line

The following rules and time-line apply for all tests for EBN111/122 as illustrated by the figure below:

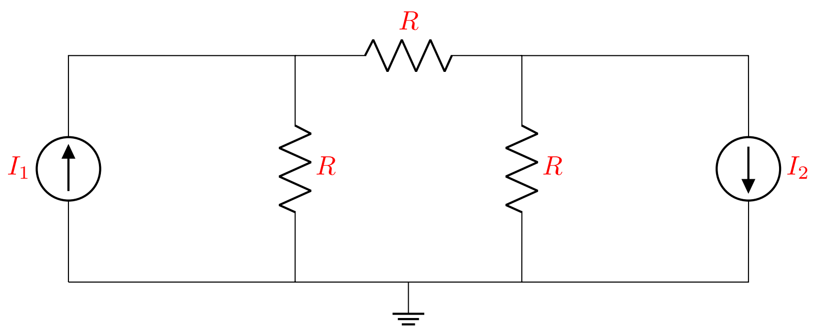
1. The Download Grace Period (DGP) is a 15 minute period before the official start of the test for students to download all documents related to the test and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the test. Students must work through the test at an average rate of at least 2 minutes per mark.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the test.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.ams@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final test mark.
8. No late submissions is accepted after the PP.



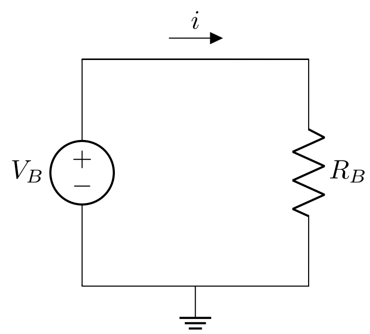
Section 1

Question 1 [2]	
Given: I_1, I_2, R.	
Use source transformation to obtain Circuit B so that it is equivalent to Circuit A, and determine i in Circuit B.	

Circuit A:



Circuit B:

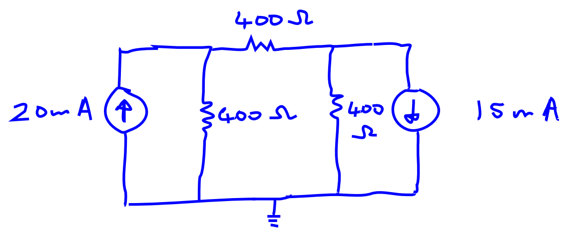


01	i [2]

* Given: $I_1 = 20\,mA$

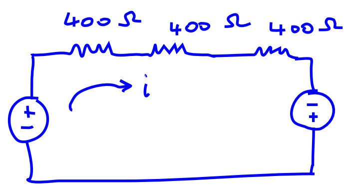
$I_2 = 15\,mA$

$R = 400\,\Omega$



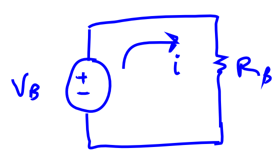
Source transformation:

$V_1 = 20\,mA \times 400\,\Omega = 8\,V$



$V_2 = 15\,mA \times 400\,\Omega = 6\,V$

$\Rightarrow$

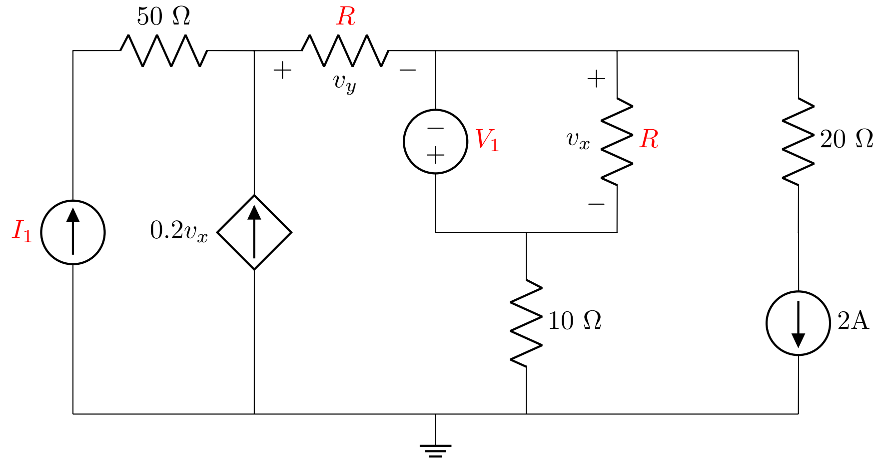


$i = \frac{V_B}{R_B}$ and $V_B = 8\,V + 6\,V = 14\,V$

$R_B = 3 \times 400\,\Omega = 1.2\,k\Omega$

$\therefore i = \frac{14\,V}{1.2\,k\Omega} = 11.667\,mA$

Question 2 [2]
Given: I_1, V_1, R.
Determine $v_y^{I_1}$, the independent contribution of I_1 to v_y.



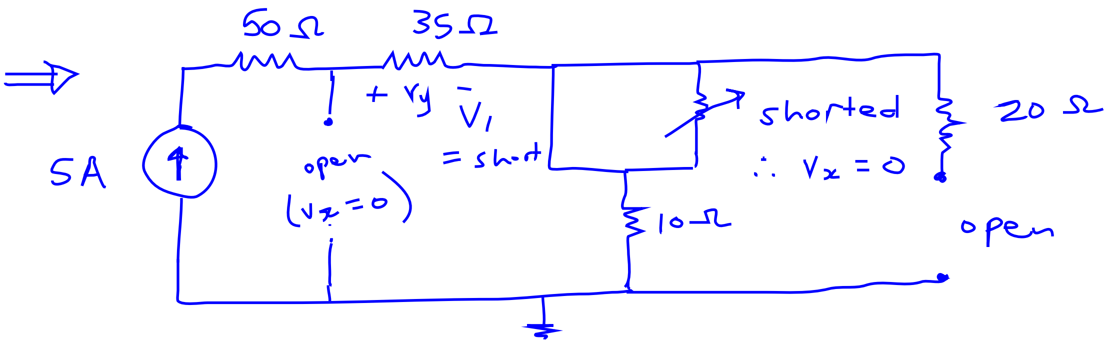
02	$v_y^{I_1}$ [2]

* Given: $I_1 = 5$ A

$V_1 = 8$ V

$R = 35\ \Omega$

⟹



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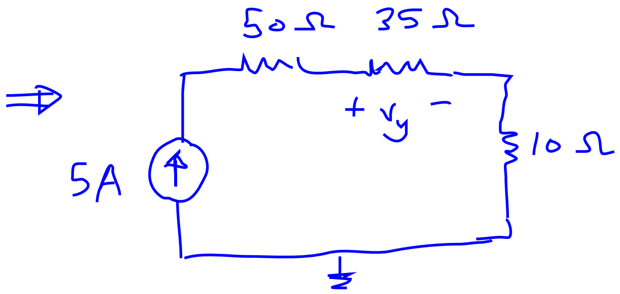
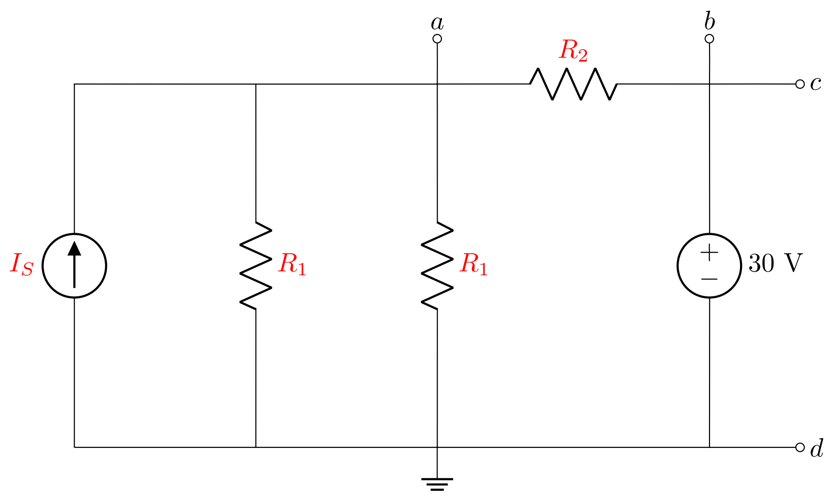


$$v_y^{5A} = IR$$

$$v_y^{5A} = 5A \times 35\ \Omega$$

$$v_y^{5A} = 175\ \text{V}$$

Questions 3 to 4 [4]	
Given: I_S, R_1, R_2.	
Determine the Thevenin equivalent voltage V_{Th} and resistance R_{Th} between terminals a-b.	



03	V_{Th} [2]
04	R_{Th} [2]

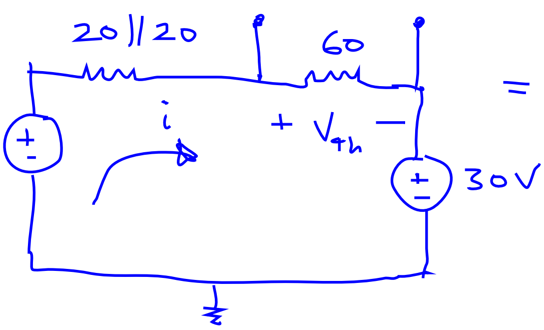
∗ Given: $I_S = 5\,A$

$R_1 = 20\,\Omega$

$R_2 = 60\,\Omega$

V_{Th} : source transformation

$\Rightarrow$



$20 \| 20 = 10\,\Omega$

$V_s = 5A \times (20 \| 20)$

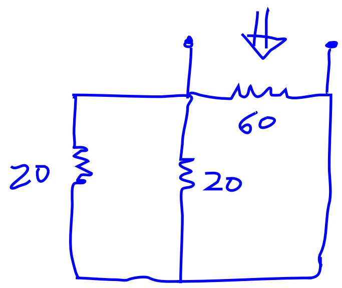
KVL: $-50V + 70\,i + 30\,V = 0$

$i = \frac{20}{70} = 285.714\text{ mA}$

$V_{Th} = 60\,\Omega \times i = 17.143\text{ V}$

$\Downarrow$

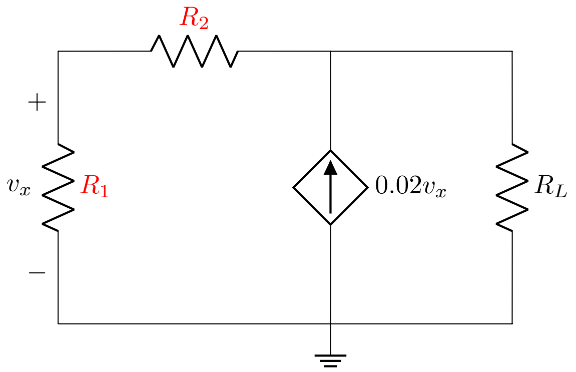
R_{Th} :



$R_{Th} = 20 \| 20 \| 60$

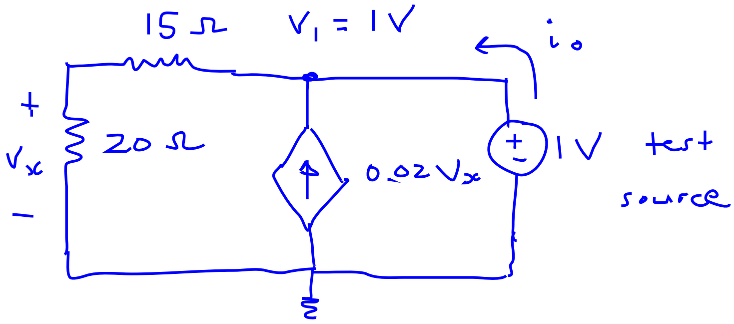
$= 10 \| 60 = \frac{600}{70}$

$\therefore R_{Th} = 8.5714\,\Omega$

Question 5 [2]	
Given: R_1, R_2.	
Determine the value of the load resistor R_L for maximum power transfer to R_L.	
	
05	R_L [2]

* Given : $R_1 = 20\,\Omega$

$R_2 = 15\,\Omega$



$R_L = R_{Th}$ for max power

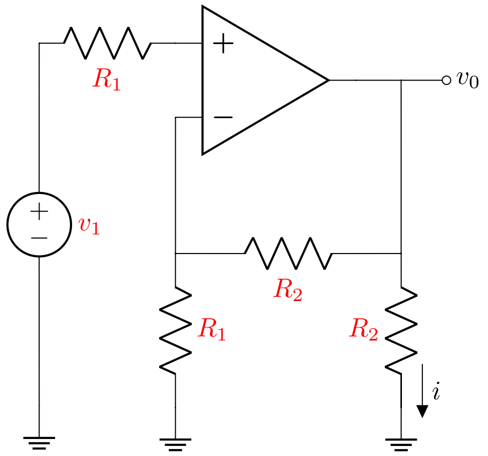
$V_1 = 1V$

$$V_x = V_1\left(\frac{20}{20+15}\right) = \frac{20}{35}$$

$$\text{KCL:} \quad i_o = \frac{1V}{20\Omega + 15\Omega} - 0.02V_x = \frac{1V}{35\Omega} - 0.02\left(\frac{20}{35}\right)$$

$$\therefore i_o = 28.571\,mA - 11.4285\,mA = 17.1425\,mA$$

$$R_L = R_{TH} = \frac{1V}{i_o} = \frac{1}{17.1425\,mA} = \underline{58.334\,\Omega}$$

Question 6 [2]	
Given: v_1, R_1, R_2.	
Assume that the op-amp is ideal. Determine the current i.	
	
06	i [2]

* Given: $v_1 = 7V$

$R_1 = 1k\Omega$

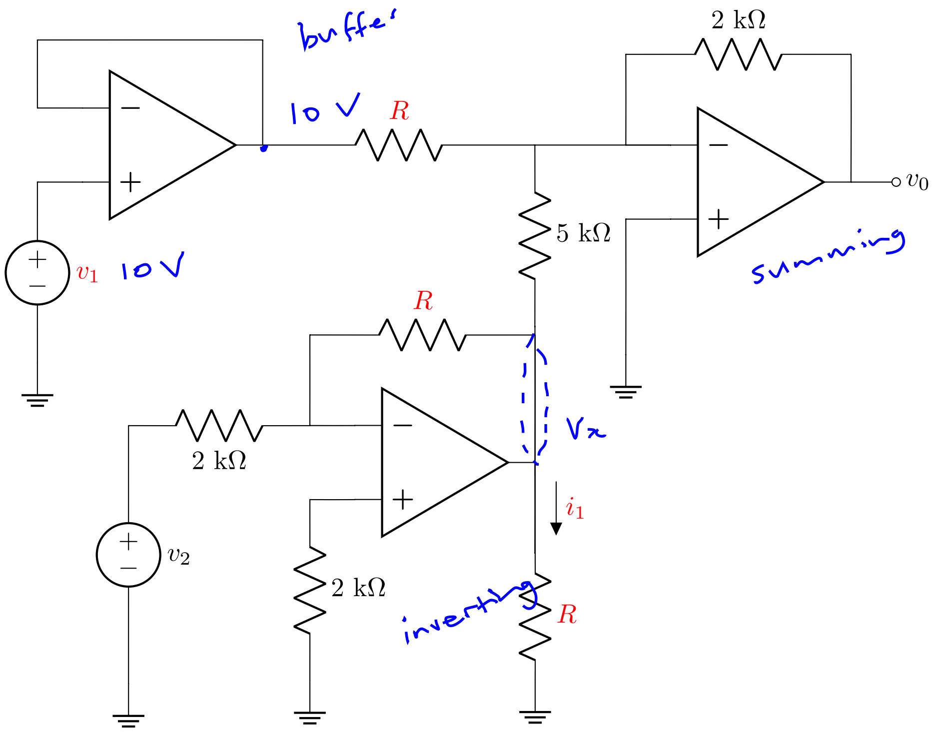
$R_2 = 4k\Omega$

non-inverting : $V_o = 7V\left(1 + \frac{4k\Omega}{1k\Omega}\right)$

$\therefore V_o = 7(5) = 35V$

$i = \frac{V_o}{4k\Omega} = \frac{35V}{4k\Omega} = 8.75mA$

Questions 7 to 8 [4]	
Given: i_1, v_1, R.	
Assume that the op-amps are ideal. Determine the voltage source v_2 and the output voltage v_0.	
07	v_2 [2]
08	v_0 [2]



* Given: $i_1 = 1\,mA$

$V_1 = 10\,V$

$R = 7\,k\Omega$

$\underline{V_2}$: inverting configuration:

$\therefore V_x = V_2\left(-\frac{7k\Omega}{2k\Omega}\right)$

$V_2 = V_x\left(-\frac{2k\Omega}{7k\Omega}\right)$

and $V_x = i_1 R = 1\,mA \times 7k\Omega = 7V$

$\therefore V_2 = 7\left(\frac{-2}{7}\right) = -2\,V$

$\underline{V_o}$: summing configuration:

$\therefore V_o = -\left(\frac{2k}{7k}(10) + \frac{2k}{5k}\underbrace{V_x}_{7V}\right)$

$\therefore V_o = -(2.8571 + 2.8)$

$\therefore V_o = -5.6571\,V$

Total Section 1: [16]

Grand Total : [16]